ATRACURIUM

High-alert drug

Organ-independent elimination

CHEMICAL NAME 2,2′-{1,5-pentanediylbis[oxy(3-oxo-3,1-propanediyl)]}bis[1-(3,4-dimethoxybenzyl)-6,7-dimethoxy-2-methyl-1,2,3,4-tetrahydroisoquinolinium] dibesilate

CLASS Non-depolarising neuromuscular blocker (benzylisoquinolinium), intermediate-acting

TRADE NAMES Tracrium · Atracurium besilate injection

THERAPEUTIC USES / INDICATIONS

- Muscle relaxation for intubation and surgery.
- Facilitation of mechanical ventilation in ICU.
- Useful in renal or hepatic failure (organ-independent breakdown).

ADMINISTRATION

- **IV** bolus or infusion; only with ventilation and sedation.
- Store in fridge (2–8 °C).
- Do not mix with alkaline solutions (e.g. thiopental).

DOSAGES

- **Intubation (adult/child >1 month):** 0.3–0.6 mg/kg.
- **Maintenance:** 0.1–0.2 mg/kg as needed, or infusion 0.3–0.6 mg/kg/h.
- Monitor with nerve stimulator (train-of-four).

PHARMACOKINETICS

- **Onset:** 2–3 min. **Duration:** 20–35 min.
- Eliminated by Hofmann degradation (spontaneous at body pH/temperature) and ester hydrolysis.
- Metabolite laudanosine (CNS stimulant at high levels). **t½:** 20 min.

PHARMACODYNAMICS

- Competitive antagonist at **nicotinic ACh receptors** of motor end-plate → flaccid paralysis.
- Reversed by anticholinesterases (neostigmine).
- Causes histamine release.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Flushing, skin wheals (histamine release).
- Hypotension, tachycardia.

ADVERSE EFFECTS (SERIOUS)

- Bronchospasm; anaphylaxis.
- Awareness under paralysis if sedation inadequate.
- Prolonged weakness (ICU myopathy) with long infusions.
- Seizures from laudanosine accumulation (rare).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- Inability to ventilate.
- **Caution:** asthma (histamine), myasthenia gravis (very sensitive), hypothermia (slower breakdown).

KEY INTERACTIONS

- **Volatile anaesthetics, aminoglycosides, magnesium, clindamycin:** potentiate block.
- **Phenytoin, carbamazepine (chronic):** resistance.

NURSING MANAGEMENT

- Never give without sedation and airway control.
- Monitor train-of-four, SpO_2, ventilator parameters.
- Watch for flushing, hypotension, bronchospasm.
- ICU: eye care (taping/lubricant), pressure-area care, sedation assessment.
- Make sure reversal drugs available at end of surgery.

PATIENT EDUCATION

- Explain to family that the patient cannot move but may hear; speak calmly and reassuringly.

KEY POINTS

- Non-depolarising blocker, intermediate duration.
- Histamine release → flushing, hypotension.
- Hofmann elimination — safe in renal/liver failure.
- Reversed by neostigmine.

SOURCES CHECKED **[90]** Atracurium Besilate 10 mg/ml Solution for Injection/Infusion (UK SmPC) **[91]** Atracurium Besylate, IV (US label) **[92]** Atracurium Besylate, IV (US label, AuroMedics Pharma)

Full references p. 432–457